October 18, 2026

Editor-in-Chief
IEICE Transactions on Information and Systems

Dear Editor-in-Chief,

I am submitting the enclosed manuscript, "Locating and Mitigating Factual Hallucination Circuits in Small Language Models via Activation Patching," for consideration as a Regular Paper in the IEICE Transactions on Information and Systems.

This work presents a systematic activation-patching study of context–memory conflict in three small language models: Gemma-2-2B, Llama-3.2-1B, and Qwen2.5-1.5B. The main contributions are:

1. **Causal localization.** A coarse-to-fine patching map shows that restoration effects concentrate in late-layer attention outputs at the answer position, replicating across all three architectures (Spearman 0.81–1.00).
2. **Head characterization.** Bidirectional patching with a neutral-control analysis identifies a single specialist head (L22.H4) whose neutral effect is minimal in both bfloat16 and float32, and separates it from dual-role heads and from a bfloat16-selected exploratory set.
3. **Quantified intervention trade-offs.** Single-head knockout of the specialist reduces distractor-following by 35% while increasing first-token accuracy by 5.3 points, at MMLU retention 0.992 and a 5% perplexity cost. All results are evaluated on held-out relations, with fact-clustered bootstrap CIs and exact McNemar tests.
4. **Transparent baselines.** Eight of ten random single-head knockouts outperform the specialist behaviorally at comparable capability cost; two carry severe metric-specific costs. The manuscript reports this openly rather than claiming superiority of the selection procedure.

All experimental data, analysis scripts, and per-item records are publicly available on OSF (DOI: 10.17605/OSF.IO/53MYH). The supplementary material accompanying this submission contains the complete dataset, head lists, split assignments, and environment details for reproduction.

I confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. I have no conflicts of interest to disclose.

Thank you for your consideration.

Sincerely,

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